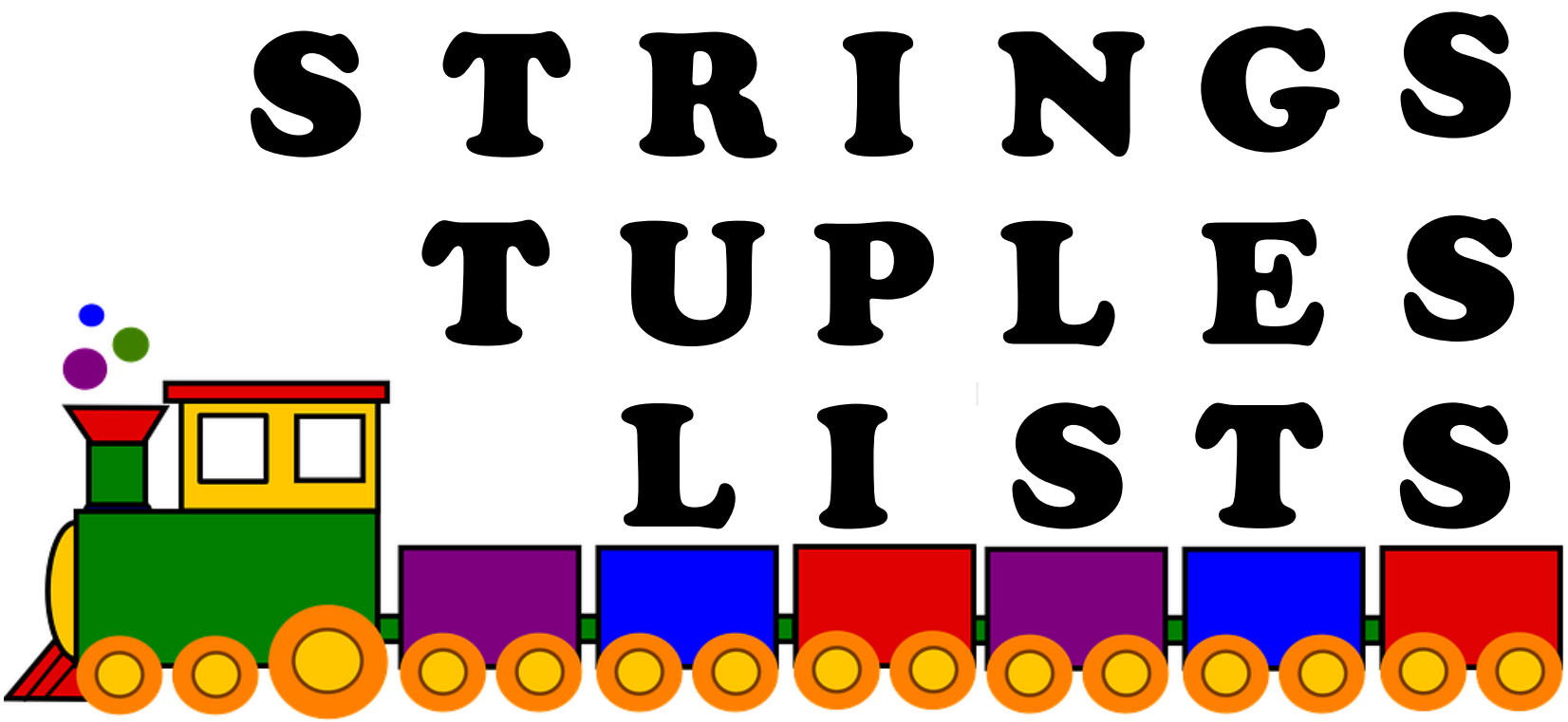
A Quick Tour of Python

Dr. Sumit Kumar Tetarave

ML Lab #1: Part 2

Programming with Python



Strings

- Strings in Python have type str
- They represent sequence of characters
  - Python does not have a type corresponding to character.
- Strings are enclosed in single quotes(') or double quotes(“)
  - Both are equivalent
- Backslash (\) is used to escape quotes and special characters

Strings

```
>>> name='intro to python'
>>> descr='acad\'s first course'
>>> name
'intro to python'
>>> descr
"acad's first course"
```

- More readable when print is used

```
>>> print descr
acad's first course
```

Length of a String

- len function gives the length of a string

```
>>> name='intro to python'
>>> empty=''
>>> single='a'
>>> len(name)
15
>>> len(single)
1
>>> len(empty)
0
>>> special='1\n2'
>>> len(special)
```

\n is a **single** character: the special character representing newline

Concatenate and Repeat

- In Python, + and * operations have special meaning when operating on strings
  - + is used for concatenation of (two) strings
  - * is used to repeat a string, an int number of time
  - Function/Operator Overloading

Concatenate and Repeat

```
>>> details = name + ', ' + descr
>>> details
"intro to python, acad's first course"

>>> print punishment
I won't fly paper airplanes in class

>>> print punishment*5
I won't fly paper airplanes in class
I won't fly paper airplanes in class
I won't fly paper airplanes in class
I won't fly paper airplanes in class
I won't fly paper airplanes in class
```

string concatenation using join() method

```
words = ["Hello", "world"]
sentence = " ".join(words)
# 'Hello world'
```

```
# Using an f-string
age = 30
greeting = f"I am {age} years old."
# 'I am 30 years old.'
```

Indexing

- Strings can be indexed
- First character has index 0

```
>>> name='Acads'
>>> name[0]
'A'
>>> name[3]
'd'
>>> 'Hello'[1]
'e'
```

Indexing

- Negative indices start counting from the right
- Negatives indices start from -1
- -1 means last, -2 second last, ...

```
>>> name='Acads'
>>> name[-1]
's'
>>> name[-5]
'A'
>>> name[-2]
'd'
```

Indexing

- Using an index that is too large or too small results in "index out of range" error

```
>>> name='Acads'
>>> name[50]

Traceback (most recent call last):
  File "<pyshell#136>", line 1, in <module>
    name[50]
IndexError: string index out of range
>>> name[-50]

Traceback (most recent call last):
  File "<pyshell#137>", line 1, in <module>
    name[-50]
IndexError: string index out of range
```

Slicing

- To obtain a substring
- s[start:end] means substring of s starting at index start and ending at index end-1
- s[0:len(s)] is same as s
- Both start and end are optional
  - If start is omitted, it defaults to 0
  - If end is omitted, it defaults to the length of string
- s[:] is same as s[0:len(s)], that is same as s

Slicing

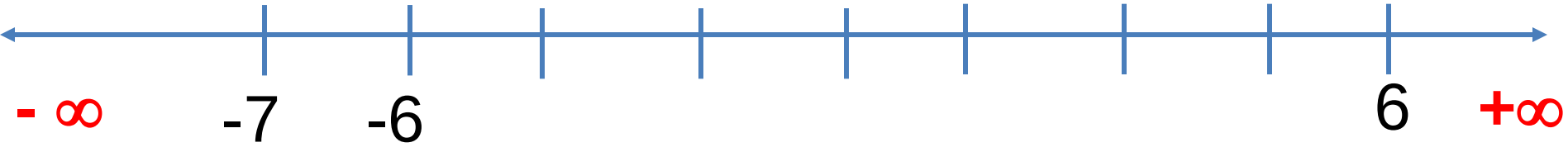
```
>>> name='Acads'
>>> name[0:3]
'Aca'
>>> name[:3]
'Aca'
>>> name[3:]
'ds'
>>> name[:3] + name[3:]
'Acads'
>>> name[0:len(name)]
'Acads'
>>> name[:]
```

More Slicing

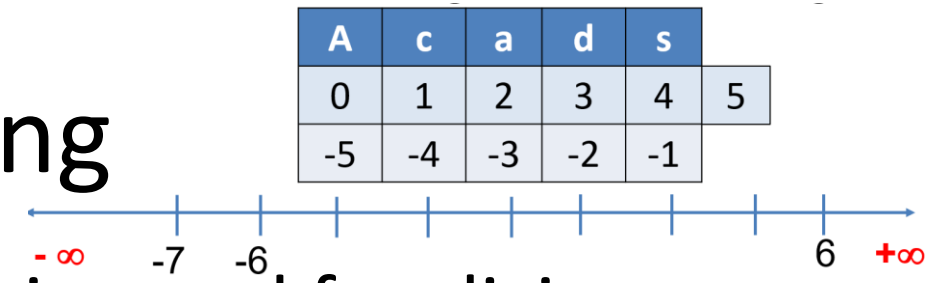
```
>>> name='Acads'
>>> name[-4:-1]
'cad'
>>> name[-4:]
'cads'
>>> name[-4:4]
'cad'
```

Understanding Indices for slicing

A	c	a	d	s	
0	1	2	3	4	5
-5	-4	-3	-2	-1	



Out of Range Slicing



- Out of range indices are ignored for slicing
- when start and end have the same sign, if start >=end, empty slice is returned

```
>>> name='Acads'
>>> name[4:50]
's'
>>> name[40:50]
''
>>> name[-50:20]
'Acads'
```

```
>>> name[-50:-20]
''
>>> name[50:20]
''
>>> name [1:-1]
'cad'
```

Why?

The step parameter allows you to include characters within the slice at regular intervals.

```
# Every second character in the string
every_second = details[::2]
print every_second
```

```
# Reversing a string using slicing
reversed_text = details[::-1]
print reversed_text
```

Tuples

- A tuple consists of a number of information separated by commas

```
>>> t = 'intro to python', 'amey karkare', 101
>>> t[0]
'intro to python'
>>> t[2]
101
>>> t
('intro to python', 'amey karkare', 101)
>>> type(t)
<type 'tuple'>
```

- Empty and Singleton Tuples

```
>>> empty = ()
>>> singleton = 1,   # Note the comma at the end
```

Nested Tuples

- Tuples can be nested

```
>>> course = 'Python', 'Amey', 101
>>> student = 'Prasanna', 34, course
>>> student
('Prasanna', 34, ('Python', 'Amey', 101))
```

- Note that course tuple is copied into student.
  - Changing course does not affect student

```
>>> course = 'Stats', 'Adam', 102
>>> student
('Prasanna', 34, ('Python', 'Amey', 101))
```

Length of a Tuple

- len function gives the length of a tuple

```
>>> course = 'Python', 'Amey', 101
>>> student = 'Prasanna', 34, course
>>> empty = ()
>>> singleton = 1,
>>> len(empty)
0
>>> len(singleton)
1
>>> len(course)
3
>>> len(student)
3
```

More Operations on Tuples

- Tuples can be concatenated, repeated, indexed and sliced

```
>>> course1
('Python', 'Amey', 101)
>>> course2
('Stats', 'Adams', 102)
>>> course1 + course2
('Python', 'Amey', 101, 'Stats', 'Adams', 102)
>>> (course1 + course2)[3]
'Stats'
>>> (course1 + course2)[2:7]
(101, 'Stats', 'Adams', 102)
>>> 2*course1
('Python', 'Amey', 101, 'Python', 'Amey', 101)
```

Unpacking Sequences

- Strings and Tuples are examples of sequences
  - Indexing, slicing, concatenation, repetition operations applicable on sequences
- Sequence Unpacking operation can be applied to sequences to get the components
  - *Multiple assignment* statement
  - LHS and RHS must have equal length

Unpacking Sequences

```
>>> student
('Prasanna', 34, ('Python', 'Amey', 101))
>>> name,roll,regdcourse=student
>>> name
'Prasanna'
>>> roll
34
>>> regdcourse
('Python', 'Amey', 101)
>>> x1,x2,x3,x4 = 'amey'
>>> print(x1,x2,x3,x4)
a m e y
```